Parity Generator and Checker

Architecture Program for Parity Generator

```
library ieee;
use ieee.std_logic_1164.all;
entity paritygen is
port (A,B,C: in std_logic;
        Ep, Op: out std_logic);
end paritygen;
architecture behave of paritygen is
begin
Ep<= (A Xor(B Xor C));
Op <= not (A Xor(B Xor C));
end behave;
```

Test bench Parity Generator:

```
library ieee;
use ieee.std_logic_1164.all;
entity paritygen_tb is
end paritygen_tb;

architecture test of paritygen_tb is
component paritygen
port (A,B,C: in std_logic;
        Ep, Op: out std_logic);
end component;
signal A,B,C,Ep,Op:std_logic;
begin
parity_gen:paritygen port map (A=>A, B=>B, C=>C, Ep=>Ep, Op=>Op);
Process begin
A<='0';
B<='0';
C<='0';
wait for 20 ns;
A<='0';
B<='0';
C<='1';
wait for 20 ns;
A<='0';
B<='1';
C<='0';
wait for 20 ns;
A<='0';
B<='1';
C<='1';
wait for 20 ns;
```

```
A<='1';
B<='0';
C<='0';
wait for 20 ns;
A<='1';
B<='0';
C<='1';
wait for 20 ns;
A<='1';
B<='1';
C<='0';
wait for 20 ns;
A<='1';
B<='1';
C<='1';
wait for 20 ns;
wait;
end process;
end test;
```

Architecture Program for Parity Checker:

```
library ieee;

use ieee.std_logic_1164.all;

entity paritycheck is

port (A,B,C,D: in std_logic;

        Epc, Opc: out std_logic);

end paritycheck;

architecture behave of paritycheck is

begin

Epc<= ((A Xor B) Xor (C Xor D));

Opc <= not ((A Xor B) Xor (C Xor D));

end behave;
```

Test bench Parity Checker:

```
library ieee;

use ieee.std_logic_1164.all;

entity paritycheck_tb is

end paritycheck_tb;


architecture test of paritycheck_tb is

component paritycheck

port (A,B,C,D: in std_logic;

        Epc, Opc: out std_logic);

end component;

signal A,B,C,D,Epc,Opc:std_logic;

begin

parity_check:paritycheck port map (A=>A, B=>B, C=>C,D=>D, Epc=>Epc, Opc=>Opc);

Process begin

A<='0';

B<='0';
```

```
C<='0';
D<='0';
wait for 1 ns;
A<='0';
B<='0';
C<='0';
D<='1';
wait for 1 ns;
A<='0';
B<='0';
C<='1';
D<='0';
wait for 1 ns;
A<='0';
B<='0';
C<='1';
D<='1';
wait for 1 ns;
A<='0';
B<='1';
C<='0';
D<='0';
wait for 1 ns;
A<='0';
B<='1';
C<='0';
D<='1';
wait for 1 ns;
A<='0';
```

```
B<='1';
C<='1';
D<='0';
wait for 1 ns;
A<='0';
B<='1';
C<='1';
D<='1';
wait for 1 ns;
A<='1';
B<='0';
C<='0';
D<='0';
wait for 1 ns;
A<='1';
B<='0';
C<='0';
D<='1';
wait for 1 ns;
A<='1';
B<='0';
C<='1';
D<='0';
wait for 1 ns;
A<='1';
B<='0';
C<='1';
D<='1';
wait for 1 ns;
```

```
A<='1';
B<='1';
C<='0';
D<='0';
wait for 1 ns;
A<='1';
B<='1';
C<='0';
D<='1';
wait for 1 ns;
A<='1';
B<='1';
C<='1';
D<='0';
wait for 1 ns;
A<='1';
B<='1';
C<='1';
D<='1';
wait for 1 ns;
wait;
end process;
end test;
```